

Lucas Monir William

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SUMMARY

Backend Software Engineer with hands-on experience building **production systems** using **NestJS, Express, Laravel, PostgreSQL, Redis, and RabbitMQ**. Experienced in designing concurrency-safe, real-time, geo-spatial tracking and event-driven backend architectures.

EXPERIENCE

Backend Developer — Bunyan Tech

Jun 2025 – Apr 2026

Alexandria, Egypt • Full-time

- Engineered production REST APIs in **NestJS** and **TypeScript** powering core business systems for **1,000+ active users**, cutting average response time through **PostgreSQL** query optimization, database indexing, and Redis caching.
- Designed and implemented concurrency control using **Mutex locks** and database transactions, **eliminating race** conditions and guaranteeing data consistency under concurrent multi-user writes.
- Built a real-time task management system using **React, Express** and **WebSockets** with role-based access and live status updates, reducing internal task-tracking overhead.

Technical Trainee — TEDx Dokki Youth

Feb 2025 – Sep 2025

Alexandria • Hybrid

- Designed and built a full-stack event booking system using **Express, React** and **PostgreSQL** processing registrations for **200+ attendees**, using transactional writes to prevent double-booking under concurrent requests.
- Automated registration and data-handling workflows for live events, replacing manual spreadsheet-based tracking during peak sign-up periods.

Software Development Intern — Prodigy InfoTech

Aug 2024 – Sep 2024

Remote • Internship

- Built a full-stack job portal platform using **React, Node.js** and **MySQL** with JWT authentication and role-based access for recruiters and workers, including REST APIs.

Backend Development Intern — Code Alpha

Jul 2024 – Aug 2024

Remote • Internship

- Developed backend services in **Node.js** for a restaurant management platform, a ticket-booking system, and a URL shortener, designing normalized MySQL schemas and transaction-safe booking APIs to prevent double-booking

PROJECTS

Egypt Supply Chain Visibility Platform | NestJS, React, PostgreSQL, Prisma, Redis, RabbitMQ, Socket.IO, Docker, CI/CD

github.com/ahmedmedhat-se/egypt-supply-chain-visibility

- Implemented real-time shipment tracking using **Socket.IO** and a self hosted OSRM routing engine, enabling live map based shipments visualization and route aware tracking.
- Designed a Redis cache layer for high-frequency dashboard reads to reduce database load, documented **20+ REST endpoints** in Swagger, and hardened the API with **Helmet, rate limiting, and signed tokens**.
- Built a **CI pipeline** that validates the application on pushes and pull requests, builds the application and **Dockerize** on version update and pushes it to **GHCR**.

Tournament Management System | PHP, MySQL, JavaScript, Blade, CI/CD, unit/integration testing

github.com/loucass/Tournament-management-system

- Built a custom PHP MVC framework from scratch (front-controller, PSR-4 autoloader, router) powering a multi-role tournament platform with live scoring on a normalized 6-table schema, hardened with CSRF tokens and prepared statements, and validated by 50 tests plus a GitHub Actions CI pipeline.

ExifWipe | Python, CI/CD, unit/integration testing

github.com/loucass/exifwipe

- Built a Python CLI for lossless EXIF/metadata removal across 15+ image/document formats (JPEG, PNG, TIFF/RAW, HEIC, PDF, etc.), using file-signature detection instead of extension trusting, and validated output integrity with a 131-test suite verified against ExifTool across 3 different python versions.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, PHP
Backend: NestJS, Express, Laravel, Node.js, Socket.IO, WebSockets
Databases: PostgreSQL, MySQL, Redis, Prisma
Testing: unit, integration
Messaging & Queue: RabbitMQ, BullMQ
API & Security: REST, Swagger, JWT, RBAC, Helmet, Rate Limiting
Frontend: React, Blade
Tools & Infra: Docker, Linux, Git, CI/CD

EDUCATION

Bachelor's degree in Computer Science — Alexandria University 2025 – Expected 2029
Faculty of Computer and Data Science
WE Applied Technology School — Programming Department Graduated 2025
High school specialized in programming, networking and communication.

ACHIEVEMENTS

- **IEEE YESIST12 Junior Einstein track (National):** ranked **#1** nationally (2024) and Top 10 (2025); qualified for international rounds in Malaysia and Tunisia.
- **Top 7 Finalist — Tech Tank Egypt (National):** pitched “WattWizard”, an industrial energy-efficiency platform, to national industry leaders including Ahmed El-Sewedy, Dina Ghabbour, and Abdallah Sallam.
- **Egypt Science & Engineering Fair (ISEF):** achieved the 3rd place nationally (2024) and Top 10 (2025).
[TV interview: [link](#)]